



legoESM: A Differentiable Earth System Model in JAX

Technical Documentation of Equations and Implementations

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Introduction

legoESM is a modular Earth System Model written entirely in JAX, supporting automatic differentiation through all components. The model couples atmospheric, oceanic, land, and sea-ice components on the sphere, with 25 dynamical cores spanning five discretisation families (centred, finite-volume PPM, pseudospectral, FC-Gram, C-grid) and over thirty physics parameterisations.

Design principles.

- Pure-JAX implementation: all components are compatible with `jit`, `grad`, and `vmap`.
- Modular physics: each parameterisation follows a factory pattern returning a callable `physics_fn(state, grid, coord) → Tendencities`.
- Multiple dynamical cores share the same physics interface, enabling interchangeable experiments.
- Neural-network components (SFNO) integrate seamlessly via the same JAX computational graph.

Grid Systems

Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection**. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be gnomonic coordinates on a face.

Face-to-Cartesian mapping

For face 0 (+x face):

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y), \quad (1)$$

with analogous expressions for the remaining five faces. Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, and geographic coordinates follow as $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

Cell area

The area element on the gnomonic equidistant grid is

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (2)$$

Grid spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (3)$$

Metric factors $\Delta x_{i,j}$ and $\Delta y_{i,j}$ are computed as two-cell spans for second-order centred differences.

Grid angle and wind rotation

The angle θ between the grid x -axis and geographic east is

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (4)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (5)$$

Halo exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back using the padded grid angle.

Two halo widths are supported. **Halo = 1** (one ghost cell per side) suffices for second-order centred differences and MUSCL reconstruction. **Halo = 2** (two ghost cells per side) is required by higher-order reconstruction stencils (PPM, WENO5). The halo exchange interpolates neighbour-face strips to the correct gnomonic position at each depth, removing the $O(\Delta\alpha)$ mismatch near cube corners.

Gaussian Grid and Spectral Transforms

The spectral dynamical cores use a Gaussian grid with triangular truncation at total wavenumber $n_{\max}$.

Grid dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\max} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\max} + 1)(n_{\max} + 2)}{2}. \quad (6)$$

Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced.

Associated Legendre polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. They are computed via the recursion

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (7)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (8)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

Spherical harmonic transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (9)$$

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\max}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (10)$$

Spectral operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

Velocity from vorticity and divergence

Streamfunction $\psi = \nabla^{-2} \zeta$, velocity potential $\chi = \nabla^{-2} \delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (11)$$

where θ is colatitude.

Sigma Vertical Coordinate

$$\sigma = p / p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (12)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$. Layer thickness $\Delta\sigma_k = \sigma_{k+1/2} - \sigma_{k-1/2}$.

Simmons–Burrige geopotential

The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (13)$$

where the Simmons–Burrige coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta\sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (14)$$

Sigma-dot

The vertical velocity in σ -space is obtained by integrating the continuity equation from the model top ($\dot{\sigma} = 0$):

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}. \quad (15)$$

Pressure velocity

Since $p = \sigma p_s$, the material pressure tendency is

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}, \quad (16)$$

where dp_s/dt is the *material* derivative of surface pressure (Eulerian tendency plus advection by the horizontal wind).

Height and z^* Coordinates

The terrain-following z^* coordinate:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (17)$$

Physical height: $z(z^*, x, y) = z_s + z^* (H - z_s)/H$.

For the **ocean**, the z^* coordinate has a dynamic Jacobian:

$$J(\eta) = \frac{\eta + H}{H_{\max}}, \quad (18)$$

where η is the free-surface elevation and H the resting depth.

Core Operators

All finite-difference operators use second-order centred differences on the A-grid (collocated) with halo padding on the cubed sphere. The discrete difference operators δ_i and δ_j are defined as centred differences of *cell-centre* values on the padded grid:

$$\delta_i[q]_{i,j} = q_{i+1,j} - q_{i-1,j}, \quad \delta_j[q]_{i,j} = q_{i,j+1} - q_{i,j-1}. \quad (19)$$

All metric-weighted fluxes ($u h_y$, $v h_x$) are evaluated at cell centres before differencing, consistent with the A-grid stencil.

Gradient

$$\left. \frac{\partial f}{\partial x} \right|_{i,j} = \frac{f_{i+1,j} - f_{i-1,j}}{2 \Delta x_{i,j}}, \quad \left. \frac{\partial f}{\partial y} \right|_{i,j} = \frac{f_{i,j+1} - f_{i,j-1}}{2 \Delta y_{i,j}}. \quad (20)$$

Divergence (flux form)

$$\nabla \cdot (u, v) = \frac{1}{2 A_{i,j}} \left[\delta_i(u h_y) + \delta_j(v h_x) \right], \quad (21)$$

where $h_x = \Delta x/2$, $h_y = \Delta y/2$ are half-edge lengths and A is the cell area.

Vorticity

$$\zeta = \frac{1}{2 A_{i,j}} \left[\delta_i(v h_y) - \delta_j(u h_x) \right]. \quad (22)$$

Laplacian and Hyperdiffusion

$$\nabla^2 f = \nabla \cdot (\nabla f), \quad \text{Hyperdiffusion:} \quad -\nu \nabla^4 f = -\nu \nabla^2 (\nabla^2 f). \quad (23)$$

Finite-Volume Transport

The finite-volume dynamical cores use upwind reconstruction at cell edges coupled with a Rusanov approximate Riemann solver. Three reconstruction methods are available, selected via the **limiter** parameter.

MUSCL Reconstruction (2nd order, halo = 1)

Piecewise-linear reconstruction with slope limiting (van Leer, 1979). At the interface between cells i and $i+1$:

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (24)$$

where ϕ_i is the limited slope. Two limiters are provided:

- **Minmod**: $\phi = \minmod(\Delta q_+, \Delta q_-)$,
- **MC** (monotonised central): $\phi = \minmod(\minmod(2\Delta q_+, 2\Delta q_-), \frac{1}{2}(\Delta q_+ + \Delta q_-))$.

PPM Reconstruction (4th order, halo = 2)

The Piecewise Parabolic Method (Colella & Woodward, 1984) constructs a monotone parabola in each cell from two edge values and the cell average.

Fourth-order edge values

At the interface $i + \frac{1}{2}$:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (25)$$

Monotonicity constraint

The edge value is clamped to lie between its two adjacent cell averages:

$$\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1}). \quad (26)$$

Colella–Woodward limiting

For each cell i with left edge value $a_L = \hat{q}_{i-1/2}$ and right edge value $a_R = \hat{q}_{i+1/2}$:

1. If $(a_R - q_i)(q_i - a_L) \leq 0$ (local extremum): set $a_L = a_R = q_i$.
2. Define $\delta_m = a_R - a_L$ and $\delta_6 = 6(q_i - \frac{1}{2}(a_L + a_R))$.
 - If $\delta_m \delta_6 > \delta_m^2$: set $a_L = 3q_i - 2a_R$.
 - If $\delta_m \delta_6 < -\delta_m^2$: set $a_R = 3q_i - 2a_L$.

The left and right states at each edge are then $q_L = a_R$ of the cell to the left and $q_R = a_L$ of the cell to the right.

WENO5 Reconstruction (5th order, halo = 2)

The 5th-order Weighted Essentially Non-Oscillatory scheme (Jiang & Shu, 1996) combines three 3rd-order sub-stencil polynomials with nonlinear weights that adapt to local smoothness.

Left-biased reconstruction at $i + \frac{1}{2}$

Using cells $q_{i-2}, \dots, q_{i+2}$:

$$q^{(0)} = \frac{1}{6}(2q_{i-2} - 7q_{i-1} + 11q_i), \quad d_0 = \frac{1}{10}, \quad (27)$$

$$q^{(1)} = \frac{1}{6}(-q_{i-1} + 5q_i + 2q_{i+1}), \quad d_1 = \frac{6}{10}, \quad (28)$$

$$q^{(2)} = \frac{1}{6}(2q_i + 5q_{i+1} - q_{i+2}), \quad d_2 = \frac{3}{10}. \quad (29)$$

Smoothness indicators

$$\beta_0 = \frac{13}{12}(q_{i-2} - 2q_{i-1} + q_i)^2 + \frac{1}{4}(q_{i-2} - 4q_{i-1} + 3q_i)^2, \quad (30)$$

$$\beta_1 = \frac{13}{12}(q_{i-1} - 2q_i + q_{i+1})^2 + \frac{1}{4}(q_{i-1} - q_{i+1})^2, \quad (31)$$

$$\beta_2 = \frac{13}{12}(q_i - 2q_{i+1} + q_{i+2})^2 + \frac{1}{4}(3q_i - 4q_{i+1} + q_{i+2})^2. \quad (32)$$

Nonlinear weights

$$\alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad q_L = \sum_k \omega_k q^{(k)}, \quad \varepsilon = 10^{-6}. \quad (33)$$

Right-biased reconstruction

The right state q_R at $i + \frac{1}{2}$ is obtained by mirroring the stencil around the interface, using cells $q_{i-1}, \dots, q_{i+3}$ with ideal weights $d_0 = \frac{1}{10}$, $d_1 = \frac{6}{10}$, $d_2 = \frac{3}{10}$ applied to the reversed sub-stencils.

Rusanov Flux

At each edge, the mass flux is

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (34)$$

where $c = \sqrt{gh}$ is the gravity-wave speed (set to zero for pure transport without wave-speed dissipation). The net flux divergence for cell i is

$$-(\nabla \cdot \mathbf{F})_i = -\frac{1}{A_i} \left[(F_x h_y)_{i+1/2} - (F_x h_y)_{i-1/2} + (G_y h_x)_{j+1/2} - (G_y h_x)_{j-1/2} \right], \quad (35)$$

where h_x, h_y are the edge metric factors.

Well-balanced scheme

When topography h_s is present, the scheme reconstructs the free surface $\eta = h + h_s$ instead of h . The fluid depth at each edge is recovered as $h_L = \max(\eta_L - h_{s,\text{edge}}, 0)$, ensuring zero spurious flux for a lake-at-rest state (Audusse et al., 2004).

FC-Gram Spectral Operators

The Fourier Continuation (FC-Gram) method extends periodic Fourier spectral methods to non-periodic domains by appending a smooth continuation interval. On each cubed-sphere face, the 1D function $f(x)$ on $[0, 1]$ is extended to a periodic function on $[0, 1 + d]$ via a Gram polynomial basis in the continuation region.

Extension operator

Given f sampled at n equispaced points, the extension f_{ext} is constructed as

$$f_{\text{ext}}(x) = \sum_{k=0}^{K-1} \alpha_k G_k(x), \quad x \in [1, 1 + d], \quad (36)$$

where G_k are orthogonal Gram polynomials and the coefficients α_k are chosen so that f_{ext} smoothly matches f at both endpoints, yielding a globally smooth periodic extension.

Spectral differentiation

On the extended periodic domain of length $L_{\text{ext}} = 1 + d$, differentiation is performed via FFT:

$$f'_j = \text{IFFT} \left[i k \text{FFT}[f_{\text{ext}}] \right] \Big|_{x_j \in [0, 1]}, \quad (37)$$

where k are the Fourier wavenumbers on the extended domain. This achieves spectral accuracy in the interior with controlled Gibbs artifacts at face boundaries.

C-grid-style divergence damping

The FC-Gram variants with divergence damping add 2nd and 4th order selective dissipation of divergent modes:

$$D_{\text{div}} = -\nu_2 \nabla(\nabla \cdot \mathbf{v}) - \nu_4 \nabla(\nabla^2(\nabla \cdot \mathbf{v})), \quad (38)$$

applied to the momentum equations. This selectively damps gravity-wave modes while preserving rotational (balanced) flow, mirroring the divergence damping strategy used in FV3 on its C-D grid.

Atmospheric Dynamics

Shallow Water Equations

In **vector-invariant form** on the cubed sphere:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (39)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (40)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (41)$$

where the Bernoulli function is $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ and D_u, D_v represent hyperdiffusion.

The mass equation (39) is discretised using the finite-volume Rusanov flux (Section 5) with a configurable reconstruction method: MUSCL (2nd order), PPM (4th order), or WENO5 (5th order), selected via `ShallowWaterConfig.fv_limiter`. Momentum uses centred differences with velocity-only biharmonic hyperdiffusion.

Hydrostatic Primitive Equations (σ -coordinates)

Prognostic variables: u, v (winds), T (temperature), p_s (surface pressure).

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (42)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (43)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (44)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (45)$$

where $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, and $\kappa = R_d/c_p$.

Spectral Primitive Equations (Vorticity–Divergence)

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972):

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{[(\zeta + f)\mathbf{v}]}] + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\zeta}, \quad (46)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{[(\zeta + f)\mathbf{v}]}] - \nabla^2 (K + \Phi + \widehat{R_d T \ln p_s}) + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\delta}, \quad (47)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T\mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \widehat{\partial T / \partial \sigma} + \widehat{\kappa T \omega / p}, \quad (48)$$

$$\frac{\partial \widehat{\ln p_s}}{\partial t} = - \int_0^1 \hat{\delta} \, d\sigma. \quad (49)$$

The spectral curl and divergence of flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$ are

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (50)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices.

Non-Hydrostatic Compressible Euler Equations

In the z^* terrain-following coordinate, the prognostic variables are u, v, w, θ', ρ' (perturbations from a reference state ρ_0, θ_0, π_0).

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (51)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (52)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (53)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (54)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (55)$$

where $r(z)$ is the sponge-layer Rayleigh damping coefficient and the Exner function perturbation is

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (56)$$

Split-explicit time integration

The equations are integrated with an outer SSP-RK3 scheme for the slow tendencies (advection, Coriolis, horizontal pressure gradient, diffusion) and inner forward–backward acoustic substeps for the fast variables (w, θ', ρ') .

Forward step (w).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (57)$$

Backward step (ρ', θ').

$$\rho'^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta'^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (58)$$

An optional **semi-implicit acoustic** solve replaces the forward Euler for w with a tridiagonal system to remove the vertical acoustic CFL constraint.

Sponge layer

$$r(z) = r_{\max} \sin^2\left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}}\right). \quad (59)$$

Atmospheric Physics

Gray Radiation (Frierson 2006)

Longwave optical depth

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \tau_{\text{moist}} q_v, \quad (60)$$

with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 1150$.

Two-stream LW fluxes

Layer transmittance and emissivity:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (61)$$

Upward sweep from surface ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$):

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (62)$$

Downward sweep from TOA ($F_0^\downarrow = 0$):

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (63)$$

Shortwave (Beer–Lambert)

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-D \tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}) e^{-D \tau_{\text{SW},0} (1 - \sigma_k)}, \quad (64)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1360 \text{ W m}^{-2}$, $\alpha_s = 0.31$.

Heating rate

Defining $F_{\text{net}}^\uparrow = F^\uparrow - F^\downarrow$ (upward positive) at each interface, the heating rate for layer k is

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^\uparrow - F_{\text{net},k-1/2}^\uparrow}{p_{k+1/2} - p_{k-1/2}}, \quad (65)$$

where $k + \frac{1}{2}$ is the bottom interface and $k - \frac{1}{2}$ the top (pressure increasing downward). Positive divergence of upward flux warms the layer.

RRTMGP Correlated- k Radiation

The RRTMGP backend calls JAX-RRTMGP with correlated- k gaseous optics (128 LW and 112 SW g -points), a two-stream RTE solver, and optional cloud optics. Default trace-gas concentrations: CO₂=415 ppmv, CH₄=1900 ppbv, N₂O=332 ppbv.

SBM Convection (Frierson 2007)

Moist adiabatic lapse rate

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}, \quad (66)$$

integrated upward via a trapezoidal predictor–corrector.

Saturation mixing ratio (Tetens)

$$e_s = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d / R_v \approx 0.622. \quad (67)$$

CAPE and trigger

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (68)$$

A smooth sigmoid trigger activates convection:

$$\mathcal{F} = \sigma(\kappa_s (\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (69)$$

Enthalpy-conserving relaxation

The reference profile T_{ref} is obtained by adjusting the moist adiabat so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p (T_{\text{ref},k} - T_k) + L_v (q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (70)$$

Newton iteration (2 steps) solves for the temperature offset. Tendencies:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (71)$$

Precipitation

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (72)$$

Held–Suarez Forcing

Equilibrium temperature

$$T_{\text{eq}} = \max\left(200, \left[315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi\right] (p/p_0)^\kappa\right), \quad (73)$$

with $\Delta T_y = 60 \text{ K}$, $\Delta \theta_z = 10 \text{ K}$.

Newtonian relaxation

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right) \cos^4 \phi, \quad (74)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction

$$\frac{d\mathbf{v}}{dt} = -k_v(\sigma) \mathbf{v}, \quad k_v = k_f \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right), \quad k_f = 1/(1 \text{ day}). \quad (75)$$

Louis (1979) Turbulence

Mixing length

$$l = \frac{\kappa_{\text{vK}} z}{1 + \kappa_{\text{vK}} z / l_{\text{max}}}, \quad \kappa_{\text{vK}} = 0.4, \quad l_{\text{max}} = 100 \text{ m}. \quad (76)$$

Richardson number

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2. \quad (77)$$

Stability functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{ Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{ Ri} / \sqrt{1 + d \text{ Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (78)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

Vertical Diffusion (Implicit Thomas Algorithm)

The backward-Euler system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm.

Tridiagonal coefficients.

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (79)$$

Forward sweep.

$$w_k = \frac{a_k}{b'_{k-1}}, \quad b'_k = b_k - w_k c_{k-1}, \quad d'_k = d_k + w_k d'_{k-1}. \quad (80)$$

Back substitution.

$$\varphi_k = \frac{d'_k + c_k \varphi_{k+1}}{b'_k}. \quad (81)$$

Surface Fluxes

Three bulk flux schemes are available, selected per tile.

Constant coefficients (default)

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (82)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad (83)$$

$$\text{LH} = \rho L_v C_E |\mathbf{V}| (q_s - q_a), \quad (84)$$

with $C_D = C_H = C_E = 1.5 \times 10^{-3}$ and $|\mathbf{V}| = \sqrt{u^2 + v^2 + 10^{-4}}$.

Monin–Obukhov Similarity Theory (MOST)

Stability-dependent transfer coefficients via iterative Obukhov length computation (Businger–Dyer profiles).

Obukhov length.

$$L = -\frac{u_*^3 T_v}{\kappa_{vK} g (\theta_* + 0.61 T_a q_*)}, \quad (85)$$

where u_* , θ_* , q_* are friction velocity, temperature scale, and moisture scale.

Stability parameter. $\zeta = z_{\text{ref}}/L$.

Businger–Dyer stability functions.

$$\psi_m(\zeta) = \begin{cases} 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan x + \frac{\pi}{2}, & \zeta < 0, \quad x = (1 - 16\zeta)^{1/4}, \\ -5\zeta, & \zeta \geq 0, \end{cases} \quad (86)$$

$$\psi_h(\zeta) = \begin{cases} 2 \ln\left(\frac{1+y}{2}\right), & \zeta < 0, \quad y = (1 - 16\zeta)^{1/2}, \\ -5\zeta, & \zeta \geq 0. \end{cases} \quad (87)$$

Transfer coefficients.

$$C_D = \left[\frac{\kappa_{vK}}{\ln(z/z_0) - \psi_m(\zeta)} \right]^2, \quad C_H = \frac{\kappa_{vK}^2}{[\ln(z/z_0) - \psi_m(\zeta)][\ln(z/z_{0t}) - \psi_h(\zeta)]}. \quad (88)$$

Iterative solver. The system u_* , θ_* , q_* , z_0 , z_{0t} , z_{0q} , L is solved iteratively (default 5 iterations) using `jax.lax.fori_loop` for full JAX differentiability.

COARE 3.0 roughness (Fairall et al. 2003)

Over ocean, roughness lengths are updated each iteration:

$$z_0 = \alpha_c \frac{u_*^2}{g} + 0.11 \frac{\nu}{u_*}, \quad z_{0t} = \min(1.15 \times 10^{-4}, 5.5 \times 10^{-5} \text{Re}_*^{-0.6}), \quad (89)$$

where $\alpha_c = 0.011$ is the Charnock parameter and $\text{Re}_* = u_* z_0 / \nu$.

Large & Yeager (2004) roughness

Empirical neutral drag coefficient as a function of neutral 10-m wind speed U_{10N} :

$$C_{DN} \times 10^3 = \frac{2.7}{U_{10N}} + 0.142 + 0.0764 U_{10N}, \quad (90)$$

with roughness recovered as $z_0 = 10 / \exp(\kappa_{vK} / \sqrt{C_{DN}})$. Stanton and Dalton numbers follow the stability-corrected LY04 parameterisation.

Ocean Dynamics

Boussinesq Hydrostatic Primitive Equations

In vector-invariant form on the cubed sphere:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (91)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (92)$$

The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies.

Tracer equations (skew-symmetric)

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (93)$$

Free-surface equation

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (94)$$

Barotropic Solver (Forward–Backward, Semi-Implicit Coriolis)

The barotropic system is solved with split-explicit forward–backward substeps:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (95)$$

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (96)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$.

Equation of State (Wright 1997)

$$\rho = \frac{p + p_0(T, S)}{\lambda(T, S) + \alpha_0(T, S)(p + p_0)}, \quad (97)$$

where

$$\alpha_0 = a_0 + a_1 T + a_2 S, \quad (98)$$

$$p_0 = (b_0 + b_4 S) + T(b_1 + T(b_2 + b_3 T) + b_5 S), \quad (99)$$

$$\lambda = (c_0 + c_4 S) + T(c_1 + T(c_2 + c_3 T) + c_5 S). \quad (100)$$

Coefficients follow the MOM6 implementation. The density perturbation is $\rho' = \rho(T, S, p) - \rho_0$.

Hydrostatic Pressure and Vertical Velocity

Baroclinic pressure by downward integration:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (101)$$

Vertical velocity diagnosed bottom-up from $w(\text{floor}) = 0$:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (102)$$

Time Integration

SSP-RK3 (Shu–Osher 1988)

The three-stage strong-stability-preserving Runge–Kutta method:

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (103)$$

$$\mathbf{u}^{(2)} = \frac{3}{4} \mathbf{u}^n + \frac{1}{4} \mathbf{u}^{(1)} + \frac{1}{4} \Delta t F(\mathbf{u}^{(1)}), \quad (104)$$

$$\mathbf{u}^{n+1} = \frac{1}{3} \mathbf{u}^n + \frac{2}{3} \mathbf{u}^{(2)} + \frac{2}{3} \Delta t F(\mathbf{u}^{(2)}). \quad (105)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

Split-Explicit Methods

Each outer RK stage computes slow tendencies once, updates u, v and tracers, then runs N_s forward–backward acoustic substeps on the fast variables (w, θ', ρ') with substep size $\Delta t_s = \Delta t / N_s$.

Land, Ice, and Coupler

Slab Land Model

Energy balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) SW^\downarrow + \varepsilon_l (LW^\downarrow - \sigma_{\text{SB}} T_s^4) - SH - LH. \quad (106)$$

Bucket hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\text{max}}]. \quad (107)$$

Moisture availability $\beta = \beta_{\text{min}} + (1 - \beta_{\text{min}}) W / W_{\text{max}}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

Surface humidity consistency. All slab surface tiles (land, sea ice, lake) recompute q_{sfc} from the *updated* surface temperature T_s^{n+1} (and updated W^{n+1} for land) before constructing the `TileResponse`. This ensures that the returned q_{sfc} and T_{surface} are thermodynamically consistent at the end of the timestep.

Sea-Ice Model

Surface temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i(T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\text{min}}). \quad (108)$$

Growth and melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (109)$$

Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1-c)/h_{\text{new}} & \text{growing,} \\ (dh/dt)c/h_{\text{eff}} & \text{melting.} \end{cases} \quad (110)$$

Tile Coupler

Surface fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (111)$$

Fluxes are time-averaged over each coupling window before being passed to the atmosphere.

Ocean albedo. The constant ocean albedo α_{ocn} is controlled by `CouplerConfig.ocean_albedo` (default 0.06). When `ocean_albedo_config.method="zenith"`, the Briegleb (1992) zenith-angle formula is used:

$$\alpha_{\text{ocn}} = \frac{0.026}{1.1 \mu_0^{1.7} + 0.065} + 0.15 (\mu_0 - 0.1)(\mu_0 - 0.5)(\mu_0 - 1.0), \quad (112)$$

where $\mu_0 = \cos \theta_z$ is the cosine of the solar zenith angle.

External forcing. Time-varying greenhouse gases (CO_2 , CH_4 , N_2O), total solar irradiance, and monthly zonal-mean ozone and aerosol climatologies can be read from NetCDF files and linearly interpolated in time via the `ExternalForcingConfig` interface. Ozone and aerosol data are loaded but not yet connected to the radiation solver.

Machine Learning: Spherical Fourier Neural Operator

The SFNO (ACE/FourCastNet v2 style) operates on the Gaussian grid.

Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. **N processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** $\text{output} = \text{input} + \text{decoder}(\dots)$.

Spectral convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (113)$$

where $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair.

Conservation Corrections

Atmosphere

Dry-air mass is corrected by a uniform adjustment to p_s that restores the area-weighted global integral:

$$\Delta \bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - \sum_i A_i p_{s,i}^{\text{old}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta \bar{p}, \quad (114)$$

so that $\sum_i A_i p_{s,i}^{\text{corr}} = \sum_i A_i p_{s,i}^{\text{old}}$ (global dry mass conserved exactly). Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$.

Ocean

Volume conservation: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt conservation: uniform additive correction per unit volume.

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
R_d	Dry-air gas constant	287.04	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air heat capacity	1005.0	$\text{J kg}^{-1} \text{K}^{-1}$
R_v	Water-vapour gas constant	461.5	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
κ	R_d/c_p	0.285	—
p_0	Reference pressure	10^5	Pa
σ_{SB}	Stefan–Boltzmann constant	5.67×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth radius	6.371×10^6	m
ρ_0	Ocean reference density	1025	kg m^{-3}

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